INDEFINITE INTEGRATION

Formulae Sheet

Derivatives		Integration
[1] $\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$, $n \neq -1$	$\Rightarrow$	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$, $n \neq -1$
[2] $\frac{d}{dx} \log\lvert x\rvert = \frac{1}{x}$	$\Rightarrow$	$\int \frac{1}{x} dx = \log\lvert x\rvert + C$, $x \neq 0$
[3] $\frac{d}{dx} e^x = e^x$	$\Rightarrow$	$\int e^x dx = e^x + C$
[4] $\frac{d}{dx}\left(\frac{a^x}{\log_e a}\right) = a^x$, $a>0, a\neq 1$	$\Rightarrow$	$\int a^x dx = \frac{a^x}{\log_e a} + C$
[5] $\frac{d}{dx} \cos x = -\sin x$	$\Rightarrow$	$\int \sin x\, dx = -\cos x + C$
[6] $\frac{d}{dx} \sin x = \cos x$	$\Rightarrow$	$\int \cos x\, dx = \sin x + C$
[7] $\frac{d}{dx} \tan x = \sec^2 x$	$\Rightarrow$	$\int \sec^2 x\, dx = \tan x + C$
[8] $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$	$\Rightarrow$	$\int \operatorname{cosec}^2 x\, dx = -\cot x + C$
[9] $\frac{d}{dx} \sec x = \sec x \cdot \tan x$	$\Rightarrow$	$\int \sec x \cdot \tan x\, dx = \sec x + C$
[10] $\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cdot \cot x$	$\Rightarrow$	$\int \operatorname{cosec} x \cdot \cot x\, dx = -\operatorname{cosec} x + C$
[11] $\frac{d}{dx} \log\lvert\sin x\rvert = \cot x$	$\Rightarrow$	$\int \cot x\, dx = \log\lvert\sin x\rvert + C$
[12] $\frac{d}{dx} \log\lvert\sec x\rvert = \tan x$	$\Rightarrow$	$\int \tan x\, dx = \log\lvert\tan x\rvert + C$
[13] $\frac{d}{dx} \log\lvert\sec x + \tan x\rvert = \sec x$	$\Rightarrow$	$\int \sec x\, dx = \log\lvert\sec x + \tan x\rvert + C$
[14] $\frac{d}{dx} \log\lvert\operatorname{cosec} x - \cot x\rvert = \operatorname{cosec} x$	$\Rightarrow$	$\int \operatorname{cosec} x\, dx = \log\lvert\operatorname{cosec} x - \cot x\rvert + C$
[15] $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$, $\lvert x\rvert < 1$	$\Rightarrow$	$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$
[16] $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$,	$\Rightarrow$	$\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$
[17] $\frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$,	$\Rightarrow$	$\int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} x + C$

Indefinite integrals

Type 1

$$\int (\sin x)^{2n}\,dx \quad , \quad \int (\cos x)^{2n}\,dx \quad , \quad n \in N$$

to solve these type of question use

$$\cos 2x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$$

Type-2

$$\int \sin^3 x\,dx \quad , \quad \int \cos^3 x\,dx$$

to solve these question use

$$\sin 3x = 3\sin x - 4\sin^3 x \Rightarrow \sin^3 x = \frac{1}{4}(3\sin x - \sin 3x)$$

$$\cos 3x = 4\cos^3 x - 3\cos x \Rightarrow \cos^3 x = \frac{1}{4}(\cos 3x + 3\cos x)$$

Type-3

$$\int \sin(mx)\cos(nx)\,dx \quad , \quad \int \sin(mx)\cdot\sin(nx)\,dx \quad , \quad \int \cos(mx)\cdot\cos(nx)\,dx$$

then $\times$ and $\div$ by 2 and use the following formula.

$$2\sin A\cdot\sin B = \cos(A-B) - \cos(A+B)$$

$$2\cos A\cdot\cos B = \cos(A+B) + \cos(A-B)$$

$$2\sin A\cdot\cos B = \sin(A+B) + \sin(A-B)$$

$\Rightarrow$ $\int F[g(x)]\,g'(x)\,dx$ then

Let $g(x) = t$

$g'(x)\,dx = dt$

$I = \int f(t)\,dt$

$I = F(t)$

$I = F(g(x))$

$\Rightarrow \int (f(x))^n\, f'(x)\,dx$ then

Let $f(x) = t$

$f'(x)\,dx = dt$

$I = \int t^n\,dt$

$I = \frac{t^{n+1}}{n+1}$

$I = \frac{(f(x))^{n+1}}{n+1}$

⇒ ⇊⇈ $\int \frac{f'(x)}{f(x)} dx$ then

Let $f(x)=t$

$f'(x)dx=dt$

$I=\int \frac{1}{t} dt$

$I=\log t$

$I=\log[f(x)]$

TYPE-4 $\int (\sin^m x)(\cos^n x)dx$, $m,n \in$ Natural number

then

⇒ if one of them is odd, the substitute for term of even power.

⇒ if both are odd the substitute either of then

⇒ if both are even even, then use identities.

TYPE-5 ⇒ $\int (\tan x)^m (\sec x)^{2n} dx$, $m,n \in N$

then substitute $\tan x=t$

⇒ $\int (\tan x)^{2m+1} (\sec x)^{2n+1} dx$, $m,n \in N$

the substitute $\sec x=t$

⇒ ⇊⇈ TRIGONOMETRIC SUBSTITUTIONS

Expression	Substitution
[1] a^2+x^2	$x=a\tan\theta$ / $a\cot\theta$
[2] a^2-x^2	$x=a\sin\theta$ / $a\cos\theta$
[3] x^2-a^2	$x=a\sec\theta$ / $a\operatorname{cosec}\theta$
[4] $\sqrt{\frac{a-x}{a+x}}$ or $\sqrt{\frac{a+x}{a-x}}$	$x=a\cos 2\theta$

TYPE-6 $\int (ax+b)\sqrt{px+q}\,dx$

then $(ax+b)=\lambda(px+q)+\mu$

find λ and μ

Some special integrals

[1] $\int \frac{1}{x^2+a^2}dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$

[2] $\int \frac{1}{x^2-a^2}dx = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|$

[3] $\int \frac{1}{a^2-x^2}dx = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|$

[4] $\int \frac{1}{\sqrt{a^2-x^2}}dx = \sin^{-1}\left(\frac{x}{a}\right)$

[5] $\int \frac{1}{\sqrt{a^2+x^2}}dx = \log|x+\sqrt{a^2+x^2}|$

[6] $\int \frac{1}{\sqrt{x^2-a^2}}dx = \log|x+\sqrt{x^2-a^2}|$

[7] $\int\sqrt{a^2-x^2}dx = \frac{1}{2}x\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)$

[8] $\int\sqrt{a^2+x^2}dx = \frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2}\log|x+\sqrt{a^2+x^2}|$

[9] $\int\sqrt{x^2-a^2}dx = \frac{x}{2}\sqrt{x^2-a^2} + \frac{a^2}{2}\log|x+\sqrt{x^2-a^2}|$

Type-7

$\int \frac{1}{ax^2+bx+c}dx$, $\int \frac{1}{\sqrt{ax^2+bx+c}}dx$, $\int\sqrt{ax^2+bx+c}\,dx$

to solve these question

make perfect square and use the above formula if

Type-8

$\int \frac{px+q}{ax^2+bx+c}dx$, $\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$, $\int(px+q)\sqrt{ax^2+bx+c}\,dx$

then

Step I ⇒ $px+q = A\frac{d}{dx}(ax^2+bx+c)+B$

$px+q = A(2ax+b)+B$ find A & B

Step II replace $px+q = A(2ax+b)+B$

Type-9 $\int \frac{P(x)}{Q(x)}dx$ where $P(x)$ is Polynomial degree $\geq Q(x)$

then to solve these type of question divide $P(x)$ by $Q(x)$

$P(x) = Q(x) \times$ quotient + Remainder

Indefinite integrals

Integrals of the type

$$\int \frac{ax^2+bx+c}{px^2+qx+d}\,dx \;,\; \int \frac{ax^2+bx+c}{\sqrt{px^2+qx+d}}\,dx \;,\; \int (ax^2+bx+c)\sqrt{px^2+qx+d}\,dx$$

to solve these type of question

$$ax^2+bx+c = \lambda(px^2+qx+d) + \mu\frac{d}{dx}(px^2+qx+d) + \gamma$$

find λ, μ and γ

⇒ Trigonometric integrals - [1]

$$\int \frac{1}{a\cos^2x+b\sin^2x}\,dx \;,\; \int \frac{1}{a+b\sin^2x}\,dx \;,\; \int \frac{1}{a+b\cos^2x}$$

$$\int \frac{1}{a+b\sin^2x+c\cos^2x}\,dx \;,\; \int \frac{1}{(a\sin x+b\cos x)^2}\,dx$$

to solve these type of question

→ divide by $\cos^2x$ in N^r and D^r

→ Change D^r in form of $\tan x$ and N^r in $\sec^2x$

→ Put $\tan x = t$

[2] $$\int \frac{1}{a\sin x+b\cos x}\,dx \;,\; \int \frac{1}{a+b\sin x}\,dx \;,\; \int \frac{1}{a+b\cos x}\,dx$$

$$\int \frac{1}{a\sin x+b\cos x+c}\,dx.$$

to solve these type of question

→ Put $\sin x = \frac{2\tan x/2}{1+\tan^2\frac{x}{2}}$ and $\cos x = \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}$

→ Change D^r in form of $\tan\frac{x}{2}$ and N^r in $\sec^2\frac{x}{2}$

→ Put $\tan\frac{x}{2} = t$

[3] $\int \frac{p\cos x + q\sin x + r}{a\cos x + b\sin x + c}\,dx$, $\int \frac{p\cos x + q\sin x}{a\cos x + b\sin x}\,dx$ [6]

to solve these type of question express numerator as

→ $N^{\prime} = \lambda(D^{\prime}) + \mu\frac{d}{dx}D^{\prime} + \gamma$

find λ, μ and γ

⇒ Algebraic Twins

→ $\int f\left(x+\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right)dx$, then Put $x+\frac{1}{x}=t$

→ $\int f\left(x-\frac{1}{x}\right)\left(1+\frac{1}{x^2}\right)dx$, then Put $x-\frac{1}{x}=t$

→ $\int \frac{x^2+1}{x^4+ax^2+1}dx$, $\int \frac{x^2-1}{x^4+ax^2+1}dx$ then divide $N^{\prime}$ and $D^{\prime}$ by x^2

→ $\int \frac{2x^2}{x^4+1}dx = \int \frac{x^2+1}{x^4+1}dx + \int \frac{x^2-1}{x^4+1}dx$

→ $\int \frac{2}{x^4+1}dx = \int \frac{x^2+1}{x^4+1}dx - \int \frac{x^2-1}{x^4+1}dx$

⇒ $\int \sqrt{\tan x}\,dx$ Put $\tan x = t^2$

⇒ $\int \sqrt{\cot x}\,dx$ Put $\cot x = t^2$

By Part

$$\int \underset{\text{I}}{f(x)}\cdot\underset{\text{II}}{g(x)}\,dx = \underset{\text{I}}{f(x)}\int \underset{\text{II}}{g(x)}dx - \int\left(\frac{d}{dx}\underset{\text{I}}{f(x)}\times\int\underset{\text{II}}{g(x)}dx\right)dx$$

Selection of Ist function by ILATE

I → Inverse Trigonometric function

L → Logarithmic function

A → Algebraic function

T → Trigonometric function

E → Exponential function

Definite Integrals

⇒ Let $\int f(x)dx = F(x)+C$ then

$$\int_a^b f(x)dx = \left[F(x)+C\right]_a^b = (F(b)+C)-(F(a)+C) = F(b)-F(a)$$

$$\boxed{\int_a^b f(x)\,dx = F(b)-F(a)}$$

⇒ Properties of Definite integrals

[1] The integration is independent of the change of variable

$$\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$$

[2] if the limits of definite integral are interchanged then its value changes by minus sign only.

$$\int_a^b f(x)dx = -\int_b^a f(x)\,dx$$

[3] $$\int_a^b f(x)\,dx = \int_a^c f(x)dx + \int_c^b f(x)dx \quad , \quad a<c<b$$

[4] $$\int_0^a f(x)dx = \int_0^a f(a-x)\,dx$$

[5] $$\int_a^b f(x)dx = \int_a^b f(a+b-x)\,dx$$

king

[6] $$\int_0^{2a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)dx, & \text{if } f(2a-x)=f(x) \\ 0, & \text{if } f(2a-x)=-f(x) \end{cases}$$

↓
jack

[7] $\int_{-a}^{a} f(x)\,dx = \begin{cases} 0 & \text{, if } f(x) \text{ is an odd function} \\ 2\int_{0}^{a} f(x)\,dx & \text{, if } f(x) \text{ is an even function} \end{cases}$

↓ queen

\# Even function :- a function $f(x)$ is said to be even if $\boxed{f(-x) = f(x)}$

\# Odd function :- a function $f(x)$ is said to be odd if

$$\boxed{f(-x) = -f(x)}$$

[8] if $f(x) \geq g(x)$ in $[a\ b]$ then

$$\int_{a}^{b} f(x)\,dx \geq \int_{a}^{b} g(x)\,dx$$

[9] $\int_{a}^{b} f(x)\cdot g(x)\,dx = \left[f(x)\int g(x)\,dx\right]_{a}^{b} - \int_{a}^{b}\left(\frac{d}{dx}f(x)\cdot\int g(x)\,dx\right)dx$

I fun (f(x)) II fun (g(x))

Differential eqn → (x, y and $\frac{dy}{dx}$ में दी eqn होती है)

⇒ Order ⇒ highest differential coefficient

⇒ Degree ⇒ D.E Polynomial form में हो तो जिसका order उसी की degree.

Types of D.E

(i) Variables-separable → जिस में dy के साथ y का function
dx के साथ x का function अलग कर लो

$$\boxed{\int f(x)dx = \int f(y)dy}$$

(ii) Reducible to variables-separable

$$\frac{dy}{dx} = f(ax+by+c) \quad \text{then let}$$

$$t = ax+by+c$$

$$\frac{dt}{dx} = a + b\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{b}\left(\frac{dt}{dx} - a\right)$$

$$\frac{1}{b}\left(\frac{dt}{dx} - a\right) = f(t)$$

$$\frac{dt}{dx} = b\,f(t) + a$$

$$\boxed{\int \frac{1}{b\,f(t)+a}\,dt = \int dx}$$

(iii) Homogeneous D.E

$$P(x,y)\,dx + Q(x,y)\,dy = 0$$

homogeneous होगा यदि सभी term में x और y के Powers का sum same होगा

* solve करने के लिए $y = ux$ put करो

$$\frac{dy}{dx} = x\frac{du}{dx} + u$$

** यदि $\frac{dx}{dy} = \frac{f(x,y)}{g(x,y)}$ हो तो

$x = uy$ put करो

$$\frac{dx}{dy} = u + y\frac{du}{dy}$$

(iv) Linear D.E

(a) $\frac{dy}{dx} + P(x)y = Q(x)$

dependent Variable सिर्फ 1 बार है, linear Power में

Step

(i) $I.F = e^{\int P(x)dx}$

(ii) $y \times (I.F) = \int Q(x) \times (I.F)\,dx + C$

(b) $\frac{dx}{dy} + P(y)x = Q(y)$

dependent variable है

Step

(i) $I.F = e^{\int P(y)dy}$

(ii) $x \times (I.F) = \int Q(y) \times (I.F)\,dy + C$

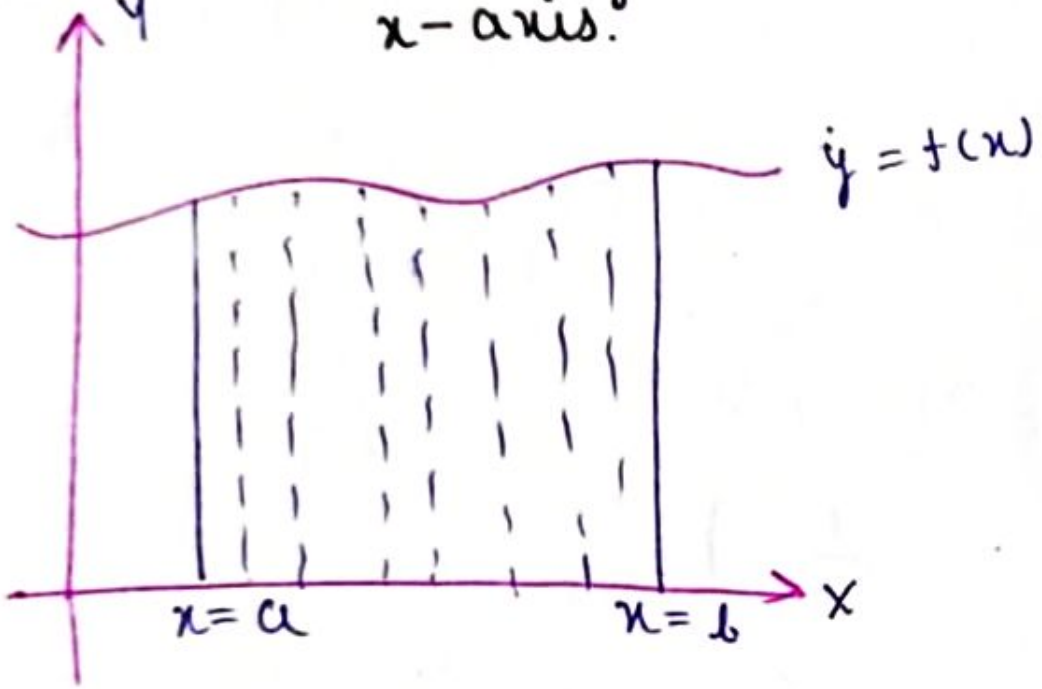
⇒ Bernoulli's Eqn

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Put $y^{1-n} = t$

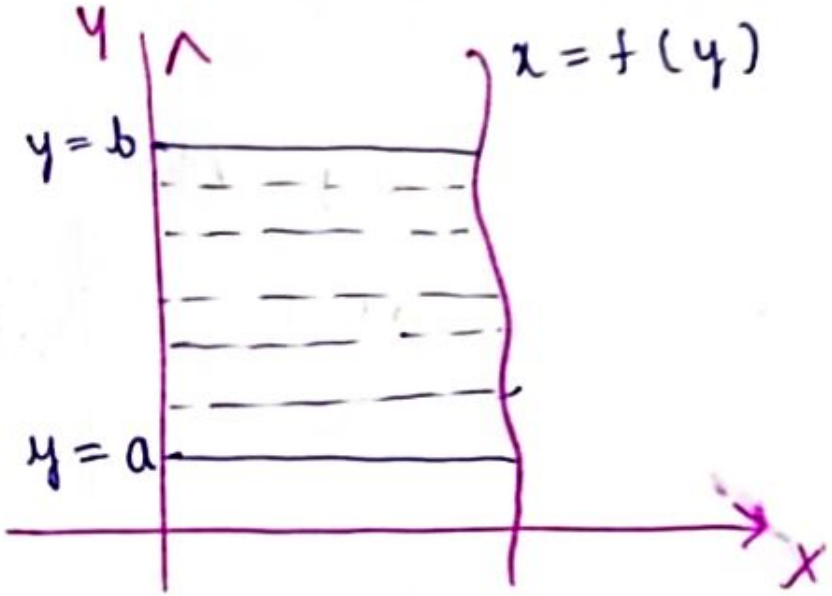
AREA BOUNDED BY CURVE

Area bounded ~~by~~ on x-axis.



$$\text{Area } a = \int_a^b y \, dx$$

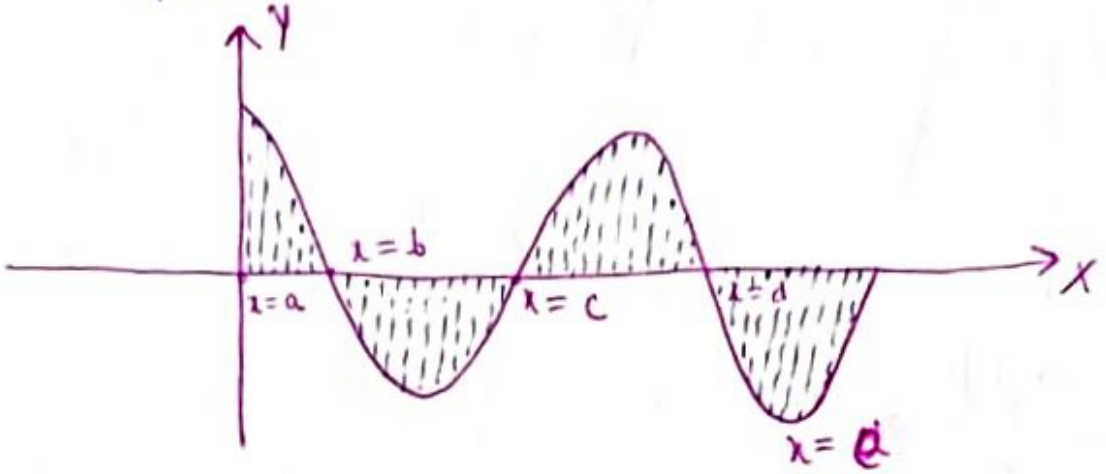
Area bounded on y-axis



$$\text{area } a = \int_a^b x \, dy$$

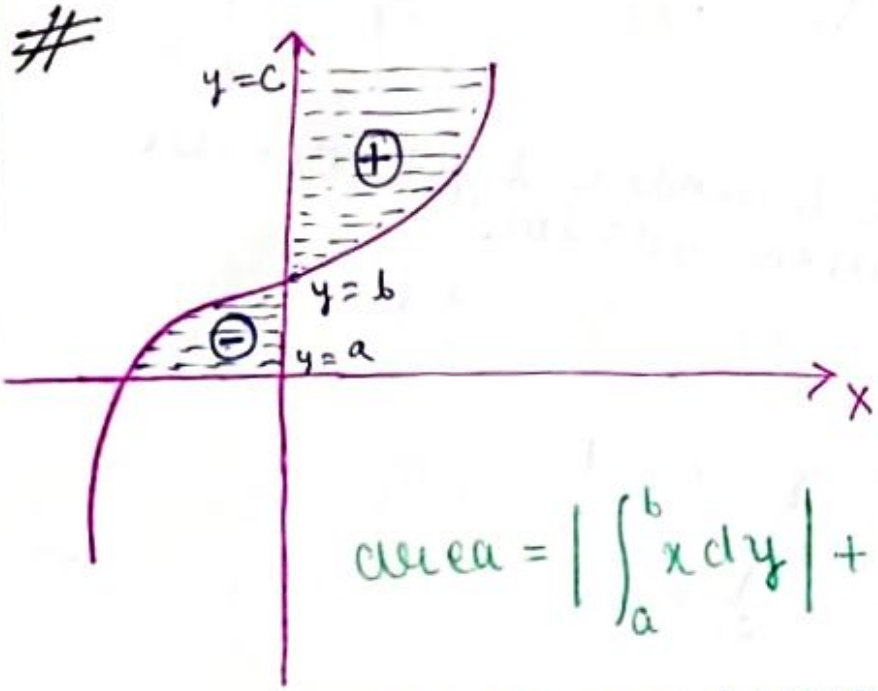
$y^2 = 4ax$

NOTE: If the area bounded by any curve in negative direction of y-axis or the x-axis, then we use mod '| |' for the total area to make negative part positive part.



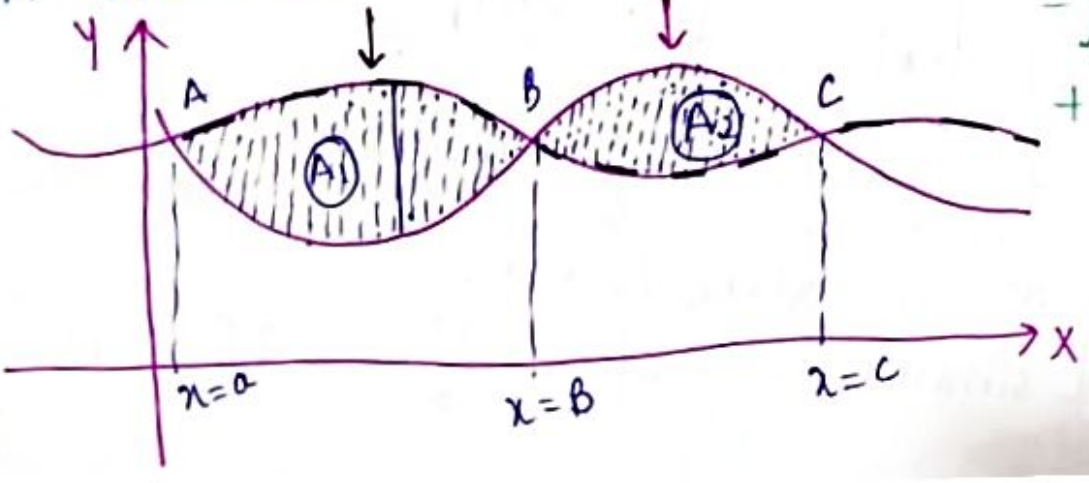
$$\text{area} = \int_a^b f(x)\,dx + \left|\int_b^c f(x)\,dx\right| + \int_c^d f(x)\,dx + \left|\int_d^e f(x)\,dx\right|$$

#



$$\text{area} = \left|\int_a^b x\,dy\right| + \int_d^c x\,dy$$

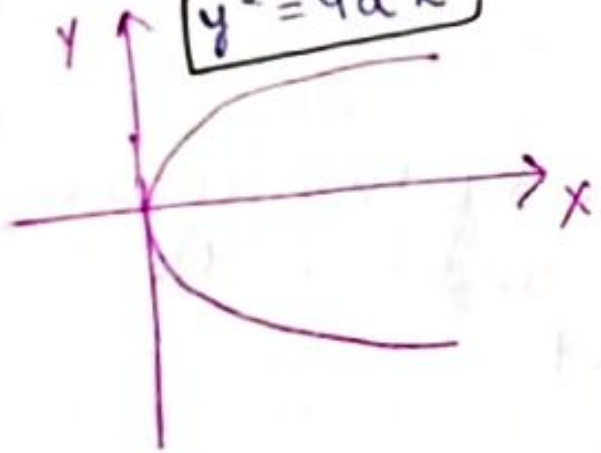
AREA BOUNDED BY TWO CURVE :



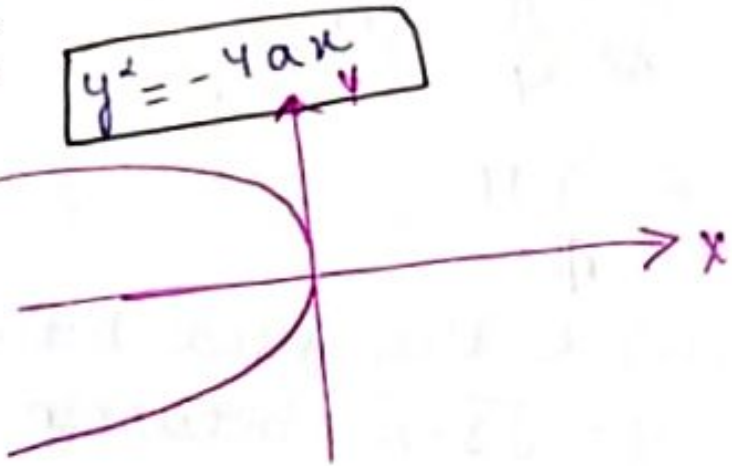
$$\text{area} = A_1 + A_2$$
$$= \int_a^b f(x) - g(x)\,dx + \int_b^c g(x) - f(x)\,dx$$

GRAPH OF PARABOLA:

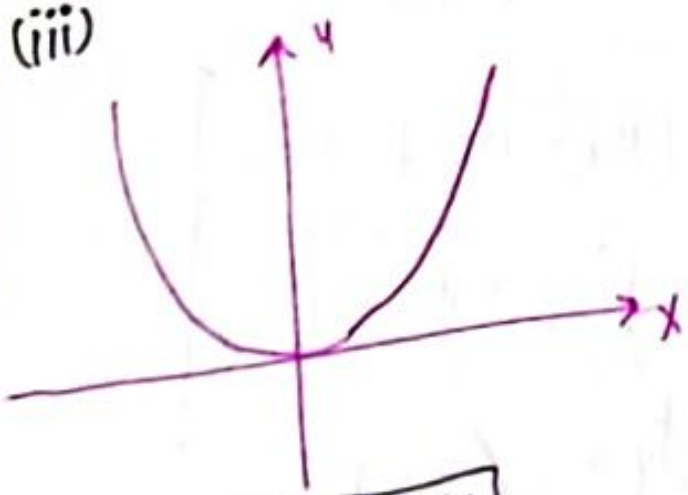
(i) $y^2 = 4ax$



(ii) $y^2 = -4ax$

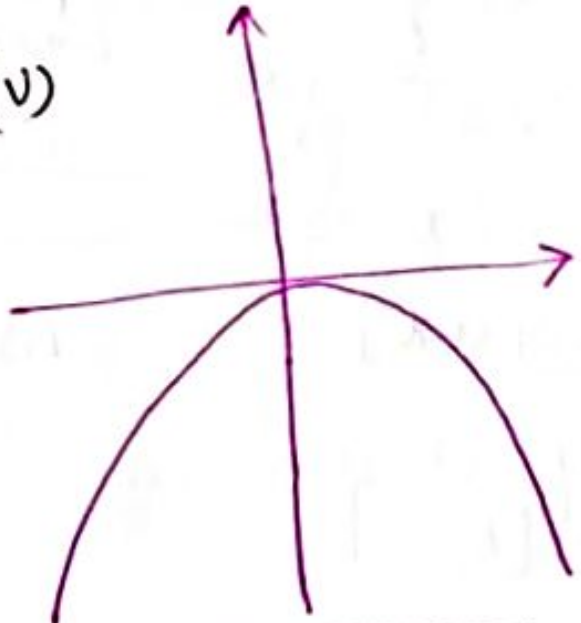


(iii)



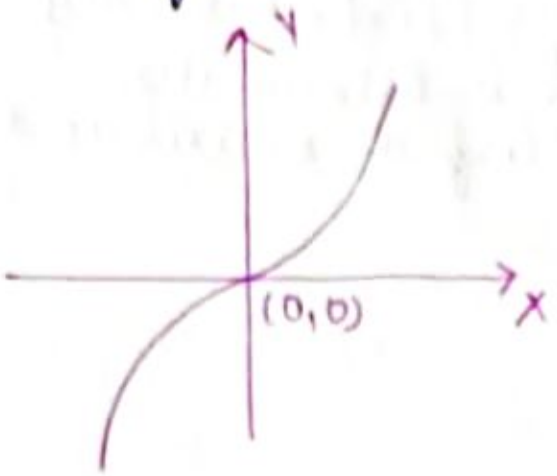
$x^2 = 4ay$

(iv)

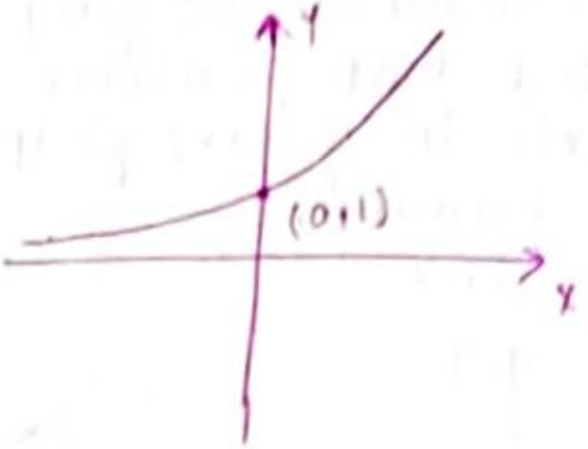


$x^2 = -4ay$

(v) $\boxed{y = x^3}$

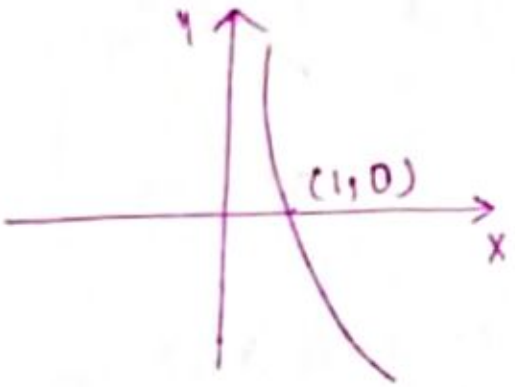


(vi) $\boxed{y = e^x}$

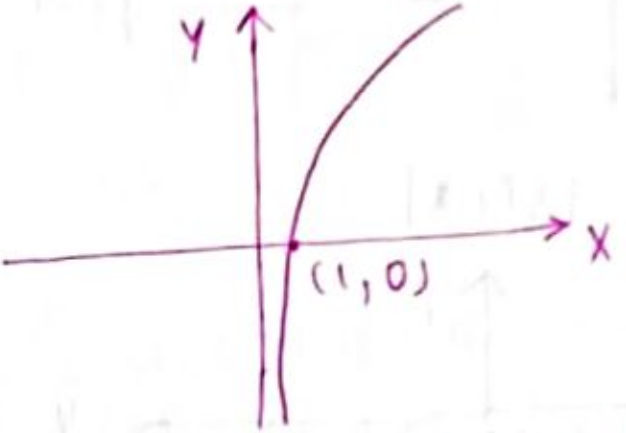


(vii) $\boxed{y = \log_a x}$

$0 < a < 1$

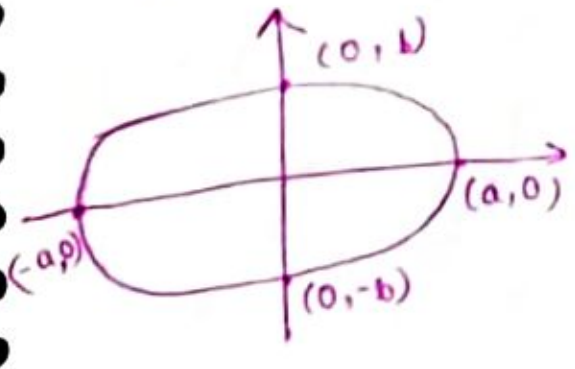


$a > 1$



#(viii) ELLIPSE : Jiska neeche ka coefficient bda graph usski axis be bnega.

$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; $\boxed{a > b}$



$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; $\boxed{b > a}$

